

Automated test summary report – No external hardware.

- The following reports are generated nightly as part of the nightly build.
- Need to create a new python script to generate a summary of the results.
- Here are the reports:
 - ~~SS-BG-SVVP-0001 - GUI (C++) Unit Test:~~ 
 - gui_test_out.txt
 - gui-coverage-report.txt
 - SS-BG-SVVP-0002 - MUX (Rust) Unit Test:
 - mux_ut_output.txt
 - mux-coverage-report.txt
 - SS-BG-SVVP-0003 - Firmware Unit Test:
 - coverage-report.txt
 - ut_output.txt
 - SS-BG-SVVP-0009 - REST Api Test (MUXomatic):
 - report-2026-05-28.txt. Note: Date is of current build and is not static.
Will always start with “report-“
 - SS-BG-SVVP-0026 - IoT Interface Test Protocol:
 - iot_integration_report.txt
- Summary of Test reports:
 - A summary of results shall be provided at the start of the report as follows:
 - ~~GUI (C++) Unit Test: PASSED -or- FAILED~~ 
 - MUX (Rust) Unit Test: PASSED -or- FAILED
 - Firmware Unit Test: PASSED -or- FAILED
 - REST Api Test: PASSED -or- FAILED
 - IoT Interface Test: PASSED -or- FAILED

 - Total results: PASSED -or- FAILED
- ~~What to read in the GUI (C++) Unit Test reports:~~
 - ~~gui_test_out:~~
 - ~~Filter out all lines that look like this:~~

~~“Totals: 18 passed, 0 failed, 0 skipped, 0 blacklisted, 60ms”~~
 - ~~There will be 20 lines to read in. Add up all the values for each line.
Review the results and see if there are any failures. The final report
shall have the following lines:~~

~~“gui_test_out results:”~~
~~“Totals: 100 passed, 0 failed, 0 skipped, 0 blacklisted”~~
~~“All tests Passed”~~

~~- or -~~
~~“There is a test Failure, <failed value> failed the test”~~

- gui-coverage-report:

- Filter out the line that looks like the following:

“TOTAL Function Coverage: 82.4% (239 of 290 functions)”

- Determine if the coverage is 80 % or higher.
- The final report shall have the following lines:

“gui-coverage-report:”

“Code Coverage: <code coverage value>”

“Code coverage Passed, <code coverage value> is >= 80%”

- or -

“Code coverage Failed, <code coverage value> is < 80%”

- What to read in the MUX (Rust) Unit Test reports:

- mux_ut_output:

- Look for a line like this:

“test result: ok. 1022 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 35.90s”

- There will other similar lines, use only this one. The final report shall have the following lines:

1 “mux_ut_output results:”

2 “test result: ok. 1022 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 35.90s”

3 “All tests Passed”

- or -

4 “There is a test Failure, <failed value> failed the test”

- mux-coverage-report:

- Filter out the line that looks like the following:

“54.96% coverage, 3841/6989 lines covered”

- Determine if the coverage is 50 % or higher.
- The final report shall have the following lines:

1 “mux-coverage-report:”

2 “Code Coverage: <code coverage value>”

3 “Code coverage Passed, <code coverage value> is >= 50%”

- or -

4 “Code coverage Failed, <code coverage value> is < 50%”

- What to read in the Firmware Unit Test reports:

- ut_output:

- Look for a line like this:

“OK (117 tests, 117 ran, 870 checks, 0 ignored, 0 filtered out, 140485 ms)”

- The final report shall have the following lines:

“ut_output results:”

“test result: ok. 117 tests, 117 ran, 870 checks, 0 ignored, 0 filtered out, 140485 ms”

“All tests Passed”

- or -

“There is a test Failure”

- coverage-report:

- List all the files under the heading “List of top 100 files that need exception reports”

- List all the files under the heading “Most changed files that need exception report”

- The final report shall have the following lines:

“coverage-report:”

“List of top 100 files that need to be included in the exception report”

“<all the files listed>”

“Most changed files that need to be included in the exception report”

“<all the files listed>”

- What to read in the REST Api Test (MUXomatic) report:

- report-YYYY-MM-DD.txt:

- Look for the last line which looks like this:

“INTEGRATAMATOR: 05/28/2026, 16:15:35: Total Failures: 0, Total Passes: 31”

- The final report shall have the following lines:

“report-YYYY-MM-DD:

“Total Failures: 0, Total Passes: 31”

“All tests Passed”

- or -

“There is a test Failure, <failed value> failed the test”

- What to read in the IoT Interface Test report:

- iot_integration_report:

- Look for the line which looks like this:

“test result: ok. 1 passed; 0 failed; 0 ignored; 0 measured; 8 filtered out; finished in 29.87s”

- The final report shall have the following lines:

“iot_integration_report:”

“1 passed; 0 failed;”

“All tests Passed”

- or -

“There is a test Failure, <failed value> failed the test”

- Example report:

Test Automation. Report Date: <date, time>

----- Test results: -----

GUI (C++) Unit Test: PASSED -or- FAILED
MUX (Rust) Unit Test: PASSED -or- FAILED
Firmware Unit Test: PASSED -or- FAILED
REST Api Test: PASSED -or- FAILED
IoT Interface Test: PASSED -or- FAILED

Total results: PASSED -or- FAILED

----- Test Summary Details: -----

GUI (C++) Unit Test:

----- gui_test_out results:

Totals: 100 passed, 0 failed, 0 skipped, 0 blacklisted
All tests Passed

gui-coverage-report:

Code Coverage: <code coverage value>
Code coverage Failed, <code coverage value> is < 80%

MUX (Rust) Unit Test:

----- mux_ut_output results:

test result: ok. 1022 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 35.90s
All tests Passed

mux-coverage-report:

Code Coverage: <code coverage value>
Code coverage Passed, <code coverage value> is >= 50%

Firmware Unit Test :

----- ut_output results:

test result: ok. 117 tests, 117 ran, 870 checks, 0 ignored, 0 filtered out, 140485 ms
All tests Passed

coverage-report:

List of top 100 files that need to be included in the exception report:

ascii.cpp 32.9137%
eeprom.cpp 66.4516%
logs.cpp 52.1073%
adc.cpp 50.9506%
app.cpp 66.6667%
vcp.cpp 51.3369%
flash.cpp 64.5503%
ascii_dbct.cpp 0%
IdleState.cpp 67.8571%
InitialFillState.cpp 0%
RestrictedIdleState.cpp 67.8571%
bus_recover.cpp 64.1791%

"Most changed files that need to be included in the exception report:

app.cpp 66.6667
logs.cpp 52.1073
adc.cpp 50.9506
ascii.cpp 32.9137
vcp.cpp 51.3369
ascii.cpp 32.9137
PollState.cpp 63.1579
flash.cpp 64.5503
RestrictedIdleState.cpp 67.8571

REST Api Test :

----- report-YYYY-MM-DD:

"Total Failures: 0, Total Passes: 31"
"All tests Passed"

IoT Integration Test :

----- iot_integration_report:

1 passed; 0 failed;
All tests Passed

